29/08/2026

A

Aakash

Medical | IIT-JEE | Foundations

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MM : 720 | **OYM - Term Test - 3 for NEET - 2028** | ***Time* : 180 Min.**

Topics Covered:

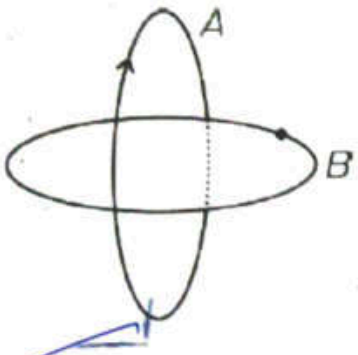
Physics	Magnetism & Matter, Electromagnetic Induction, Alternating Current, Electromagnetic Waves
Chemistry	d & f Block Elements, Coordination Compounds
Botany	Principles of Inheritance and Variation: from Linkage & Crossing Over till end and Molecular Basis of Inheritance: Up to Transcription in Prokaryotes
Zoology	Neural Control and Coordination, Chemical Coordination and Integration

General Instructions :

Duration of Test is **3** hrs.
The Test consists of **180** questions. The maximum marks are **720**.
There are three parts in the question paper consisting of **Physics, Chemistry, Botany** and **Zoology** having **45** questions in each part of equal weightage.
Each question carries **+4 marks**. For every wrong response, **-1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

Physics

1. Two coils of wires *A* and *B* are mutually at right angles to each other as shown in the figure. If current in coil *A* is changed, then in coil *B*

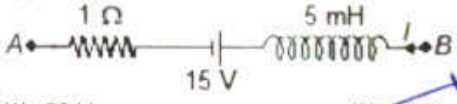


(1) No current will be induced
(2) Current will be induced in clockwise direction
(3) Current will be induced in anticlockwise direction
(4) Current will be induced depending on increasing or decreasing current

2. When a bar magnet is kept in a uniform magnetic field, the potential energy of the magnet in stable equilibrium is *x*. To keep the magnet perpendicular to the same magnetic field, the torque required is τ. Then $\frac{\tau}{x}$ is

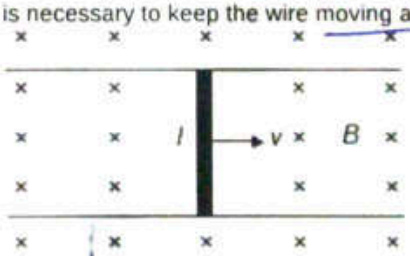
(1) 3
(2) 1
(3) $\frac{1}{2}$
(4) 2

3. The network shown in the figure is part of a complete circuit. If at a certain instant the current *I* is 5 A, and is decreasing at a rate of 10^3 A/s, then $V_B - V_A$ is (*V* : Potential)



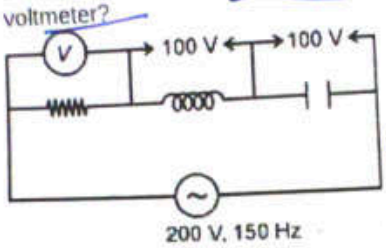
(1) 20 V
(2) 15 V
(3) 10 V
(4) 5 V

4. Figure shows a wire sliding on two parallel conducting rails placed at a separation *l*. A magnetic field *B* exists in a direction perpendicular to the plane of the rails. What force is necessary to keep the wire moving at constant velocity?



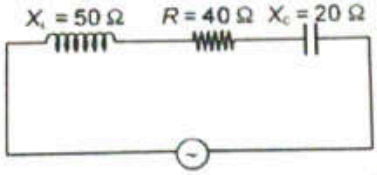
(1) $\frac{vB^2l^2}{R}$
(2) $\frac{v^2B^2l^2}{R}$
(3) $\frac{v^2l^2B}{R}$
(4) Zero

5. In the circuit given below, what will be the reading of voltmeter?



(1) 300 V
(2) 900 V
(3) 200 V
(4) 400 V

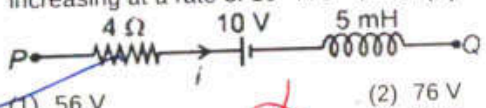
6. The power factor of the given series *RLC* circuit is



(1) 0.6
(2) 0.8
(3) 0.5
(4) 1

7. An alternating emf 100 cos(200*t*) volt is connected in series to a resistance of 20 Ω and inductance 100 mH. The phase difference between current and emf in the circuit will be
(1) $\frac{\pi}{4}$
(2) π
(3) $\frac{\pi}{2}$
(4) $\frac{\pi}{3}$

8. The network shown in the figure is a part of a complete circuit. If at a certain instant, the current *i* is 4 A and is increasing at a rate of 10^4 A s^{-1}, then $(V_P - V_Q)$ will be



(1) 56 V
(2) 76 V
(3) – 56 V
(4) 66 V

9. A : Inductance coil are made of copper.
R : Induced current is more in wire having less resistance.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

10. A square loop of edge 'a' having *n* turns is rotated with a uniform angular velocity ω about one of its diagonal which is kept fixed in horizontal position. A uniform magnetic field *B* exists in the vertical direction. E.m.f. Induced in the coil is
(1) Zero
(2) $nBa\omega \sin\omega t$
(3) $nBa\omega^2 \sin\omega t$
(4) $nBa^2\omega \sin\omega t$

11. A thin bar magnet is oscillating in earth's magnetic field in a horizontal plane with time period *T*. Now it is cut into two equal parts perpendicular to its length. Time period of oscillations of one part in same field is
(1) *T*
(2) $\frac{T}{2}$
(3) 2*T*
(4) $\frac{T}{4}$

12. Which of the following is not a vector quantity?
(1) Magnetic field
(2) Electric current
(3) Electric field
(4) Magnetic dipole momment

13. The magnetic flux linked with a coil, in weber is given by the equation $\phi = 4t^2 - t + 7$.
Then the magnitude of induced emf at 2 second will be
(1) 15 V
(2) 17 V
(3) 19 V
(4) 21 V

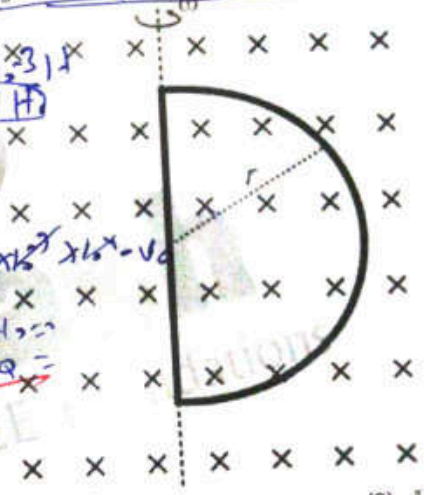
14. A flexible magnet of magnetic dipole moment *M* is bent (from its initial straight shape) into an arc subtending angle θ (in radius) at its centre of curvature. New magnetic dipole moment of the magnet is
(1) $\frac{M}{\theta}\cos\left(\frac{\theta}{2}\right)$
(2) $\frac{M}{\theta}\sin\left(\frac{\theta}{2}\right)$
(3) $\frac{2M}{\theta}\cos\left(\frac{\theta}{2}\right)$
(4) $\frac{2M}{\theta}\sin\left(\frac{\theta}{2}\right)$

15. The number of turns, cross-sectional area and length for four solenoids are given in the following table.

Solenoid	Total Turns	Area	Length
1	2N	2A	ℓ
2	2N	A	ℓ
3	3N	3A	2ℓ
4	2N	2A	$\ell/2$

The solenoid with maximum self inductance is
(1) 1
(2) 2
(3) 3
(4) 4

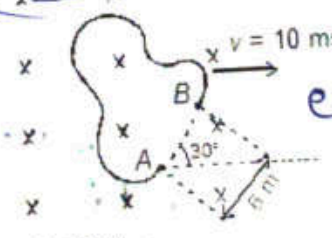
16. In a uniform magnetic field B_0 a wire in shape of a semicircle of radius *r* rotates about the diameter of the circle with angular velocity ω as shown in the figure. If the total resistance of the loop is *R*, then the mean power generated per period of revolution is



(1) $\frac{2B r^2 \omega}{\pi R}$
(2) $\frac{\pi B^2 \omega^2 r^4}{8R}$
(3) $\frac{\pi^2 B^2 \omega^2 r^4}{8R}$
(4) $\frac{\pi^2 B^2 \omega^2 r^4}{2R}$

17. A small signal voltage $V(t) = V_0 \sin\omega t$ is applied across an ideal inductor *L*. Then
(1) Current *I*(*t*) leads voltage *V*(*t*) by 90°
(2) Voltage *V*(*t*) leads current *I*(*t*) by 90°
(3) Current *I*(*t*) is in phase with *V*(*t*)
(4) Current *I*(*t*) leads voltage *V*(*t*) by 180°

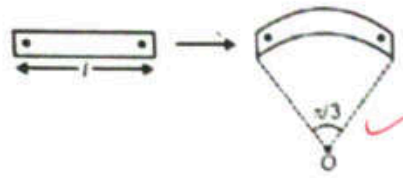
18. An irregular loop is moving with constant velocity 10 ms^{-1} in a direction perpendicular to magnetic field of magnitude 3 T and all other parameters are as arranged in figure. Induced emf across *AB* is [*AB* = 6 m as shown in figure]



(1) 70 V
(2) 90 V
(3) 45 V
(4) 75 V

19. Two coils have a mutual induction 15 mH. The current in the primary coil varies as, $I = 5\sin(100\pi t + 45°)$ mA. The peak value of voltage in the secondary coil will be
(1) 5.5π mV (2) 6.5π mV
(3) 10.5π mV (4) 7.5π mV

20. A bar magnet of length l and magnetic dipole moment M is bent as shown in figure. The new magnetic dipole moment will be

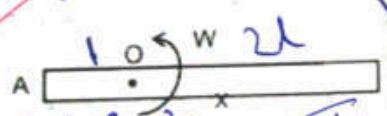


(1) M (2) $\frac{3M}{\pi}$
(3) $\frac{2M}{\pi}$ (4) $\frac{4M}{\pi}$

21. A : Magnetic field lines form closed loops.
R : Magnetic monopole does not exist.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

22. In a series LCR circuit $L = 10$ H, $C = 160$ μF and $R = 80\ \Omega$. The value of angular frequencies for which the power is half of its maximum value, are nearly
(1) 46 rad/s and 50 rad/s (2) 21 rad/s and 29 rad/s
(3) 26 rad/s and 34 rad/s (4) 26 rad/s and 30 rad/s

23. A conducting rod AC of length 3l is rotated about O in a uniform magnetic field $\vec{B}$ directed into the paper where AO = l and OC = 2l, then the magnitude of potential difference $V_A - V_C$ is : (W →Angular speed)



(1) $\frac{5wl^2B}{2}$ (2) $\frac{3wl^2B}{2}$
(3) $2Bwl^2$ (4) $\frac{Bwl^2}{2}$

24. A charged capacitor $C = 30$ μF is connected to an inductor $L = 27$ mH. The angular frequency of their oscillations is
(1) 600 rad/s (2) 1111 rad/s
(3) 1212 rad/s (4) 1700 rad/s

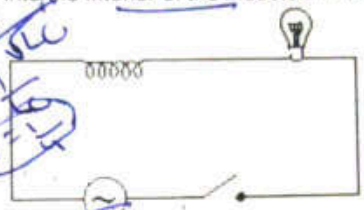
25. An AC circuit has an inductive reactance equal to 10 Ω and a resistance of 10 Ω. The effective impedance of the circuit is
(1) 20 Ω (2) $10\sqrt{2}\ \Omega$
(3) $10\sqrt{3}\ \Omega$ (4) Zero

26. The magnetic flux through a coil perpendicular to the plane is varying according to the relation $\phi = (5t^3 + 4t^2 + 2t - 5)$ Wb. Calculate the induced current through the coil at $t = 2$ s, if the resistance of the coil is 5 Ω.
(1) 10 A (2) 15.6 A
(3) 2 A (4) 61 A

27. A solenoid (air core) having 400 turns is 10 cm long and has a cross-sectional area of 0.2 cm^2. The coefficient of self inductance is (approx.)
(1) 0.4 mH (2) 40 μH
(3) 4.0 H (4) 40 H

28. A : A capacitor of suitable capacitance can be used in an AC circuit in place of choke coil.
R : A capacitor blocks DC and allows AC only
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

29. A light bulb and an inductor coil are connected to an ac source through a key as shown in the figure below. The key is closed and after sometime an iron rod is inserted into the interior of the inductor. The glow of the light bulb



(1) increases (2) decreases
(3) remains unchanged (4) will fluctuate

30. The power in AC circuit is given by $P = E_{rms}\, I_{rms} \cos\varphi$. The value of $\cos\varphi$ in series LCR circuit at resonance is
(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$
(3) 1 (4) Zero

31. If Alternating Current in a circuit is given $I = I_0\sin(100\pi t)$, then time taken by the current to change from zero to peak value will (Here I is in ampere and t is in second)
(1) $\frac{1}{200}$ s (2) $\frac{1}{50}$ s
(3) $\frac{1}{100}$ s (4) $\frac{1}{600}$ s

32. At a frequency ω_0, the reactance of a certain capacitor equal that of certain inductor. If frequency changed to $2\omega_0$, the ratio of reactance of inductor to that of the capacitor is
(1) 4 : 1 (2) $\sqrt{2}$: 1
(3) 1 : $2\sqrt{2}$ (4) 1 : 2

33. The rate of increase of current through an inductor is 2.4 A/s. If voltage drop across the inductor is 1.6 V then self inductance of the inductor will be
(1) $\frac{2}{3}$ H (2) $\frac{1}{3}$ H
(3) $\frac{3}{2}$ H (4) $\frac{1}{2}$ H

34. The ratio of rms to peak value of a sinusoidal alternating current is
(1) 1 : 2 (2) 1 : $\sqrt{3}$
(3) 1 : $\sqrt{5}$ (4) 1 : $\sqrt{2}$

35. In resonance state of LCR series circuit, resonance frequency of circuit is f. If the capacitance is made 4 times of initial value keeping inductance same, then the new resonance frequency will be

(1) $\frac{1}{4}$ times

(2) $\frac{1}{2}$ times

(3) $\frac{1}{3}$ times

(4) 4 times

36. In an LCR series circuit, at resonance

(1) The impedance is maximum

(2) Current and voltage are in phase

(3) Voltage leads the current by phase $\frac{\pi}{2}$

(4) The current is minimum

37. The power factor of an ideal choke coil is

(1) Exactly zero

(2) Nearly zero

(3) Exactly one

(4) Nearly one

38. The magnetic flux through a loop of resistance R changes as $\Delta\varphi$ in time Δt then SI unit of $\frac{\Delta\phi}{R}$ is (symbol have their usual meaning)

(1) A s

(2) $A\,s^{-1}$

(3) $J\,V^{-1}$

(4) Both (1) and (3)

39. A bar magnet of magnetic dipole moment 4 A m^2 has poles separated by 0.2 m. The magnitude of pole strength of bar magnet is

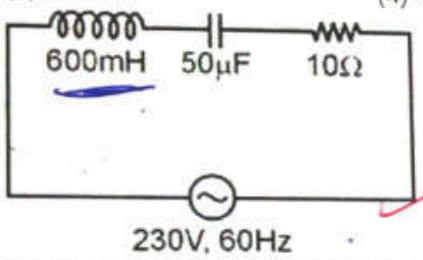
(1) 10 A m

(2) 40 A m

(3) 20 A m

(4) 30 A m

40.



The average power transferred per cycle to inductor in the given circuit is

(1) 25 W

(2) 17.42 W

(3) 16.52 W

(4) Zero

41. A transformer is employed to

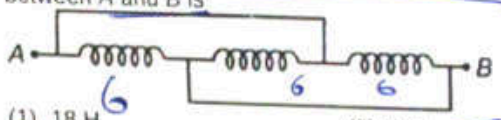
(1) Convert AC into DC

(2) Convert DC into AC

(3) Obtain a suitable AC voltage

(4) Obtain a suitable DC voltage

42. Pure inductors of inductance 6.0 H each are connected as shown in figure. The equivalent inductance of the circuit between A and B is



(1) 18 H

(2) 9 H

(3) 6 H

(4) 2 H

43. The magnetic susceptibility is positive for

(1) Diamagnetic material only

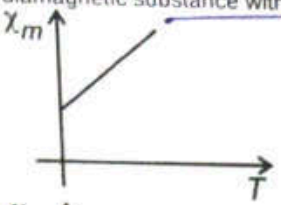
(2) Paramagnetic material only

(3) Ferromagnetic material only

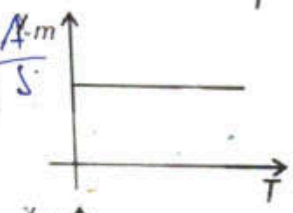
(4) Paramagnetic and ferromagnetic material

44. The graph representing variation of magnetic susceptibility of a diamagnetic substance with temperature is

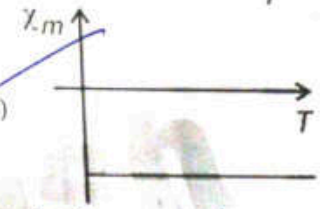
(1)



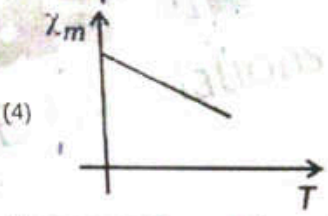
(2)



(3)



(4)



45. An ac generator converts

(1) Electrical energy into mechanical energy

(2) Mechanical energy into electrical energy

(3) Magnetic energy into mechanical energy

(4) Mechanical energy into magnetic energy

Chemistry

46. Select the incorrect statement regarding actinoids.

(1) All are silvery in appearance

(2) These are highly reactive metals

(3) HCl reacts with all actinoids

(4) All actinoids react with alkalies

47. Spin only magnetic moment of Mn^{+2} ion is

(1) $\sqrt{15}$ BM

(2) $\sqrt{35}$ BM

(3) $\sqrt{24}$ BM

(4) $\sqrt{8}$ BM

48. The correct statement(s) for the isomerism exhibited by the complexes is/are

(1) Pentacarbonylthiocyanato-S-manganese (II) ion and pentacarbonyl thiocyanato-N-manganese (II) ion are linkage isomers

(2) $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$ are solvate isomers

(3) $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$ are coordination isomers

(4) All of these

49. Effective atomic number of Cr(Z = 24) in $[Cr(H_2O)_6]Cl_3$ is

(1) 35

(2) 33

(3) 36

(4) 34

50. Among the following, the most stable complex is
(1) $[Fe(C_2O_4)_3]^{3-}$ (2) $[FeCl_6]^{3-}$
(3) $[Fe(H_2O)_6]^{3+}$ (4) $[Fe(NH_3)_6]^{3+}$

51. Number of moles of $KMnO_4$ that will be needed to react with one mole of ferrous oxalate is
(1) $\frac{3}{5}$ (2) $\frac{5}{3}$
(3) $\frac{2}{5}$ (4) $\frac{5}{2}$

52. *d*-orbital used in square planar complexes is
(1) d_{zx} (2) $d_{x^2-y^2}$
(3) d_{z^2} (4) d_{yz}

53. Select the **incorrect** statement among the following
(1) Highest oxidation state of a metal stabilizes more in its oxide than fluoride
(2) Highest oxidation state is shown by elements ruthenium and osmium.
(3) Low oxidation state of transition metals are found when a complex compound has ligands that are capable π-acceptor in addition to σ-bonding.
(4) $Cr_2O_7^{2-}$ has stronger oxidising power than MnO_4^-.

54. The hybridizations of nickel in $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ respectively are
(1) sp^3, dsp^2 (2) sp^3, sp^2
(3) dsp^2, sp^3 (4) sp^2, dsp^2

55. All halides of Cu(II) are known except the iodide because
(1) Cu^{2+} oxidises I^- to I_2
(2) Copper (II) iodide undergoes disproportionation
(3) I_2 is a good oxidizing agent
(4) Iodine atom has least electron affinity

56. Identify the incorrect statement from the following
(1) Actinoid contraction is greater from element to element than lanthanoid contraction due to more poor shielding by 5f electrons
(2) The actinoids in general show +3 oxidation state
(3) The basic strength of hydroxide of lanthanoids increases from $Ce(OH)_3$ to $Lu(OH)_3$
(4) Mischmetal consists of a lanthanoid metal (~95%) and iron (~5%) and traces of S, C, Ca and Al

57. In manufacturing of sulphuric acid the catalyst used is
(1) $TiCl_4$ (2) Fe
(3) V_2O_5 (4) $KHSO_4$

58. Maximum paramagnetism is shown by
(1) Ce^{4+} (Z = 58) (2) Yb (Z = 70)
(3) Nd (Z = 60) (4) La^{3+} (Z = 57)

59. Complex showing geometrical isomerism is
(1) $[Fe(H_2O)_6]Cl_3$ (2) $[Co(NH_3)_4Cl_2]Cl$
(3) $K_2[PtCl_4]$ (4) $[Co(en)_3]Cl_3$

60. Which of the following pair of ions is coloured in aqueous solution?
(1) Ni^{2+} and Ti^{3+} (2) Sc^{3+} and Ti^{3+}
(3) Ni^{2+} and Ti^{4+} (4) Sc^{3+} and Cu^+

61. Stability of a complex is more when
(1) Basicity of ligand is low
(2) Charge on metal ion is less
(3) Number of chelate rings are less
(4) (Charge/Size) ratio of central metal ion is high

62. Highest positive oxidation state is shown by
(1) Cl (2) Os
(3) Mn (4) Cr

63. Number of stereoisomers in Ma_3b_3 type of coordination compound is
(1) 2 (2) 3
(3) 4 (4) 8

64. Which among the following species has zero unpaired electron?
(1) Nd^{3+} (2) Gd^{3+}
(3) Lu^{3+} (4) Eu^{2+}

65. Fusion of chromite ore with sodium carbonate in presence of O_2 forms
(1) $Na_2Cr_2O_7$ (2) Cr_2O_3
(3) FeO (4) Na_2CrO_4

66. Inner orbital complex among the following is
(1) $[CoF_6]^{3-}$ (2) $[MnCl_6]^{3-}$
(3) $[Mn(CN)_6]^{3-}$ (4) $[Mn(H_2O)_6]^{2+}$

67. Which of these statements about $[Co(CN)_6]^{3-}$ is true?
(1) $[Co(CN)_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration
(2) $[Co(CN)_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration
(3) $[Co(CN)_6]^{3-}$ has four unpaired electrons and will be in a high-spin configuration
(4) $[Co(CN)_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration

68. Which of the following is not a *transuranic* element?
(1) Americium (2) Plutonium
(3) Neptunium (4) Thorium

69. What is value of CFSE for $[FeF_6]^{3-}$?
(1) $(-4 \times 2) + (+6 \times 3)$ Dq (2) $(+4 \times 3) + (-6 \times 2)$ Dq
(3) $(-4 \times 3) + (+6 \times 2)$ Dq (4) $(-4 \times 3) - (+6 \times 2)$ Dq

70. The magnetic moment of a transition metal of 3*d* series is $\sqrt{48}$ B.M. It's electronic configuration is
(1) $3d^1 4s^2$ (2) $3d^5 4s^1$
(3) $3d^6 4s^2$ (4) $3d^5 4s^2$

71. In neutral or faintly alkaline medium, $KMnO_4$ converts I^- into
(1) I_2 (2) IO_3^-
(3) IO_4^- (4) IO^-

72. Which of the following have highest magnetic moment?
(1) Cr^{2+} (2) Mn^{2+}
(3) Sc^{3+} (4) Fe^{2+}

73. Among lanthanide series, which one has largest size?
(1) Ce (2) Pm
(3) Eu (4) Gd

74. Magnetic nature and hybridization of $[Cu(NH_3)_4]^{2+}$ are
(1) Paramagnetic and sp^3 (2) Paramagnetic and dsp^2
(3) Diamagnetic and sp^3 (4) Diamagnetic and dsp^2

75. Chromate ion, CrO_4^{2-} is
(1) Diamagnetic, orange in colour
(2) Paramagnetic, yellow in colour
(3) Diamagnetic, yellow in colour
(4) Paramagnetic, orange in colour

76. Identify the incorrect statement with respect to $KMnO_4$
(1) Dark purple in colour
(2) It is decolourised by oils
(3) Paramagnetic in nature
(4) The anion has tetrahedral structure

77. Which configuration exhibits strong 'Jahn-Teller' distortion in strong ligand field?
(1) d^7 (2) d^3
(3) d^4 (4) d^5

78. Which fluoride of manganese is least likely to exist?
(1) MnF_2 (2) MnF_3
(3) MnF_4 (4) MnF_7

79. Which among the following has lowest melting point?
(1) Cr (2) Mn
(3) Fe (4) Co

80. In +2 oxidation state, the electronic configuration of a transition metal is $[Ar]3d^3$. Its atomic number is
(1) 23 (2) 24
(3) 25 (4) 26

81. Ce (Z = 58) and Yb (Z = 70) exhibit stable +4 and +2 oxidation state respectively. This is because
(1) Ce^{4+} and Yb^{2+} acquire f^7 configuration
(2) Ce^{4+} and Yb^{2+} acquire f^0 configuration
(3) Ce^{4+} and Yb^{2+} acquire f^0 and f^{14} configuration
(4) Ce^{4+} and Yb^{2+} acquire f^7 and f^{14} configuration

82. Total number of stereoisomers in $[Co(en)_2Br_2]^+$ complex are
(1) 4 (2) 3
(3) 2 (4) 6

83. The compound showing maximum number of geometrical isomers is
(1) Ma_3b_3 (2) $Ma_2b_2c_2$
(3) M(AB)(CD) (4) $M(AB)c_2d_2$

84. Enthalpy of atomization is minimum for
(1) Ti (2) Cr
(3) Mn (4) Co

Botany

91. Two genes which are very tightly linked
(1) Show very high recombination frequency
(2) Produce higher proportion of non-parental gene combinations as compared to parental ones among the progeny
(3) Are always situated on two different chromosomes
(4) Have very less distance between them on a genetic map

92. Select the **incorrect** statement w.r.t. phenylketonuria.
(1) It is an inborn error of metabolism
(2) The affected individuals lack enzyme phenylalanine hydroxylase
(3) In the affected individual, phenylalanine is converted into tyrosine
(4) Affected individuals experience mental retardation

93. Experimental verification of the chromosomal theory of inheritance was done by
(1) Sutton and Boveri (2) T.H. Morgan
(3) Punnett (4) Mendel

94. Select the **incorrect** match w.r.t. chromosomal basis of sex determination.
(1) XX-XO type – Grasshopper
(2) XX-XY type – *Drosophila*
(3) ZW-ZZ type – Butterfly
(4) ZO-ZZ type – Moth

85. A : E^0 for Mn^{3+}/Mn^{2+} is more positive than Cr^{3+}/Cr^{2+}
R : The third ionization energy of Mn is smaller than that of Cr
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

86. Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is
(1) $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(H_2O)_6]^{3+}$
(2) $[Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+}$
(3) $[Co(H_2O)_6]^{3+} < [Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+}$
(4) $[Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+} < [Co(H_2O)_6]^{3+}$

87. The number of moles of $KMnO_4$ needed to react with 1 mole of NO_2^- ion in acidic medium is
(1) 1 (2) 2
(3) $\frac{1}{5}$ (4) $\frac{2}{5}$

88. The correct statement regarding metal carbonyl, $Mn_2(CO)_{10}$ is
(1) It comprises of two square pyramidal units joined solely by a metal-metal bond
(2) It consists of bridging carbonyl group
(3) It comprises of two tetrahedral units joined by a metal-metal bond
(4) It consists of only 6 σ bonds

89. Number of secondary valencies in $CoCl_3 \cdot 6NH_3$, $CoCl_3 \cdot 5NH_3$ and $CoCl_3 \cdot 4NH_3$ respectively are
(1) 6, 5 & 4 (2) 3, 3 & 3
(3) 6, 6 & 6 (4) 3, 1 & 2

90. Which of the following complex will have highest magnetic moment?
(1) $[Mn(CN)_6]^{4-}$ (2) $[Cr(NH_3)_6]^{3+}$
(3) $[FeF_6]^{3-}$ (4) $[Cu(NH_3)_4]^{2+}$

95. Above pedigree chart is **true** for
a. X-linked recessive inheritance
b. Autosomal recessive inheritance
c. Autosomal dominant inheritance
(1) Only a (2) a and b
(3) b and c (4) Only b

96. Select the **odd** one w.r.t Mendelian disorder.
(1) Cystic fibrosis (2) Phenylketonuria
(3) Down's syndrome (4) Haemophilia

97. A normal man whose father was haemophilic, married a normal woman whose mother was haemophilic. What are the chances of getting haemophilic offsprings?
(1) 75% (2) 50%
(3) 25% (4) 100%

98. Match the Columns and select the **correct** option.

Column-I	Column-II
(A) 2n – 1	(i) Nullisomic condition
(B) 2n – 2	(ii) Trisomic condition
(C) 2n + 1	(iii) Tetrasomic condition
(D) 2n + 2	(iv) Monosomic condition

	A	B	C	D
(1)	(iv)	(i)	(ii)	(iii)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(i)	(ii)	(iii)	(iv)

(1) (1) (2) (2)
(3) (3) (4) (4)

99. Turner's syndrome
(1) Is due to extra copy of chromosome 21
(2) Is due to additional copy of the sex chromosome
(3) Leads to sterile female
(4) Is a Mendelian disorder

100. Thalassemia and sickle cell anaemia are similar as
(1) Genes responsible for both of them are present on chromosome 11
(2) Both are qualitative problem of synthesising incorrectly functioning globin protein
(3) These are X linked disorders
(4) These occur due to transversion mutation

101. In the Morgan's experiment, when he crossed yellow bodied-white eyed female *Drosophila* with the wild type male, there were 98.7% parent type individuals in F_2 generation and when white eyed-miniature winged female *Drosophila* was crossed with the wild type male, there were 62.8% parent type individuals in F_2 generation. From the given data, we can conclude that
(1) Gene for body colour and eye colour are located on non-homologous chromosomes
(2) Genes for eye colour and body colour are farther from that of genes for wing type and eye colour
(3) Crossover value for eye colour and wing type genes is more than that of body colour and eye colour genes
(4) Genes for body colour and eye colour are present only on X-chromosome but genes for wing type is present on both X and Y-chromosomes.

102. Read the following assertion and reason and select the **correct** answer from the following options.
Assertion : In sickle cell anaemia, the shape of RBC changes from biconcave disc to elongated sickle like structure.
Reason : The mutant haemoglobin molecule in sickle cell anaemia undergoes polymerization under low oxygen tension causing the change in shape of RBC.
(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

103. A : Number of linkage group is equal to the haploid set of chromosomes.
R : During linkage, the proportion of parental gene combination is much lower than the non-parental gene combinations.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

104. Recombination frequencies can be utilized in predicting the sequence of genes on the chromosome. It was suggested by
(1) T.H. Morgan (2) Mendel
(3) Alfred Sturtevant (4) Sutton and Boveri

105. Identify the following statements as true **(T)** or false **(F)** and select the **correct** option.
A. Failure of segregation of homologous chromosomes during cell division leads to aneuploidy.
B. Polyploidy is more common in plants.
C. Non-haemophilic parents cannot have a haemophilic child

	A	B	C
(1)	F	T	T
(2)	T	F	T
(3)	T	T	F
(4)	T	F	F

(1) (1) (2) (2)
(3) (3) (4) (4)

106. **Statement A**: A daughter will not normally be colour-blind unless her mother is atleast a carrier and father is colour-blind.
Statement B: A colour-blind man cannot pass the disorder to his sons but can pass on to grandsons.
Read the above statements and select the **correct** option.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A & B are incorrect
(4) Both statements A & B are correct

107. Mr. John is suffering from phenylketonuria and hypertrichosis. His father is heterozygous for phenylketonuria. The probability of John's sperm having one recessive autosomal allele and holandric gene is
(1) 1/16 (2) 1/8
(3) 1/2 (4) 1/4

108. Male honey bees
(1) Are diploid organisms
(2) Produce gametes by mitosis
(3) Are developed by fusion of male gametes with egg
(4) Do not have fathers but can produce sons

109. Complete linkage dihybrid test cross Ratio.
(1) 1 : 1 (2) 1 : 1 : 1 : 1
(3) 3 : 1 (4) 7 : 1 : 1 : 7

110. If in DNA extract of a bacterium, the proportion of cytosine was found to be 28%, then calculate the amount of thymine.
(1) 72% (2) 44%
(3) 22% (4) 56%

111. The discontinuously synthesised fragments of DNA are joined by the enzyme
(1) DNA ligase (2) DNA-dependent DNA-polymerase
(3) Phosphorylase (4) Topoisomerase

112. Select the **incorrect** match w.r.t. the structure of polynucleotide chain.

(1)	N-glycosidic linkage	A nitrogenous base is linked to the OH of 1'C of the pentose sugar
(2)	Phosphoester linkage	A phosphate group is linked to OH of 5'C of a nucleoside
(3)	Phosphodiester linkage	Two nucleotides are linked through 1'-phosphate and 5'-hydroxyl group to form a dinucleotide
(4)	H-bonds	Punnes like adenine form two hydrogen bonds with thymine from opposite strand and vice-versa

(1) (1) (2) (2)
(3) (3) (4) (4)

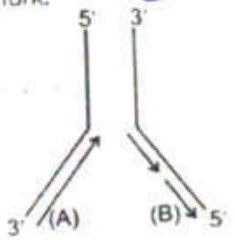
113. A hybrid DNA is allowed to replicate for one generation in a medium containing $^{15}NH_4Cl$ and for second generation in a medium containing $^{14}NH_4Cl$. What will be the percentage of heavy and hybrid DNA obtained respectively?
(1) Heavy DNA = 0%, Hybrid = 75%
(2) Heavy DNA = 75%, Hybrid = 0%
(3) Heavy DNA 25%, Hybrid = 75%
(4) Heavy DNA = 75%, Hybrid = 25%

114. During DNA replication, deoxyribonucleoside triphosphates serve dual purposes which are
(1) Acting as template and as substrate
(2) Acting as promoter and as adapter
(3) Functioning as structural molecule and as template
(4) Acting as substrate and providing energy

115. Calculate the number of nucleosomes present in the nucleus of a eukaryotic cell which possess 7.2×10^6 bp
(1) 12×10^3 (2) 36×10^3
(3) 12×10^2 (4) 36×10^2

116. Given below are two statements Assertion(A) and Reason(R). Select the **correct** option.
Assertion (A): DNA replication in *E.coli* initiates randomly at any place in DNA.
Reason (R): *E. coli* DNA lacks origin of replication site.
In the light of above statements select the **correct** option.
(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)
(2) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)
(3) (A) is true statement but (R) is false
(4) Both (A) and (R) are false statements

117. Identify the option not true for the below given replication fork.



(1) Strand A is synthesised continuously
(2) Strand B is synthesised discontinuously
(3) DNA polymerase unwind DNA helix
(4) DNA synthesis occurs in 5' to 3' direction

118. RNA and DNA both can function as genetic material but DNA is preferred over RNA because
(1) 2' – OH group in DNA is a reactive group and makes DNA labile
(2) DNA is less reactive and more stable than RNA
(3) RNA mutates at a slower rate
(4) DNA directly codes for synthesis of proteins

119. Which of the following provide(s) additional stability to the helical structure of DNA over RNA?
(a) N-glycosidic linkages
(b) Stacking of one base pair over the other in double helix
(c) Hydrogen bonding
(d) Presence of thymine instead of uracil
Choose the **correct** option
(1) (a), (b) and (c) only (2) (b) only
(3) (c) and (d) only (4) All except (a)

120. Select the **true** statement about Chargaff's rule.
(1) Molar amount of guanine is equal to that of adenine
(2) A + T / C + G is more than unity in bacteria
(3) Applicable to ssDNA
(4) A + G / T + C is equal to one

121. In the double helix structure of DNA, the pitch of the helix is _(A)_ and there are roughly _(B)_ bp in each turn.
Select the **correct** option for (A) and (B) respectively.

	(A)	(B)
(1)	3.4 A°	11 bp
(2)	0.034 nm	9 bp
(3)	0.34 nm	10 bp
(4)	3.4 nm	10 bp

(1) (1) (2) (2)
(3) (3) (4) (4)

122. Promoter in a transcription unit is located at
(1) 5 end downstream of a non-template strand
(2) 5 end downstream non-coding strand
(3) 3' end downstream of the coding strand
(4) 5 end upstream of coding strand

123. Bacteriophage $\phi \times 174$ has
(1) 48502 nucleotides (2) 48502 base pairs
(3) 5386 nucleotides (4) 5386 base pairs

124. Reverse central dogma of molecule biology can be represented by
(1) DNA → RNA → Protein (2) RNA → DNA
(3) DNA → Protein → RNA (4) Protein → DNA

125. Select **odd** one w.r.t. features of dsDNA proposed by Watson and Crick.
(1) Backbone is constituted by sugar – phosphate and bases are projected inside
(2) Two chains have anti-parallel polarity
(3) Phosphodiester linkage generates uniform distance between the two strands of the helix
(4) Two chains are coiled in a right-handed fashion

126. Which among the following histone protein is not a part of Histone octamer?
(1) H_2A (2) H_1
(3) H_2B (4) H_3 and H_4

127. A : Histone octamer has two copies of each H_2A, H_2B, H_3, H_4 and H_1.
R : In eukaryotes, positively charged DNA is wrapped around negatively charged histone proteins.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

128. A : The DNA replicates semiconservatively, the process is guided by the complementary H-bonding.
R : RNA is the first to evolve and DNA was derived from RNA.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

129. DNA replication in prokaryotes is mainly catalysed by
(1) DNA dependent RNA polymerase
(2) RNA dependent RNA polymerase
(3) RNA dependent DNA polymerase
(4) DNA dependent DNA polymerase

130. During DNA replication, unwinding of DNA double helix is catalysed by
(1) Helicase (2) Primase
(3) Ligase (4) DNA polymerase

131. A : RNA is the genetic material of QB bacteriophage.
R : RNA performs dynamic functions of messenger and adapter molecule.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

132. Select the **incorrect** match w.r.t. symbols used in pedigree analysis.
(1) ◇ – Sex unspecified
(2) ○—□ – Consanguineous mating
(3) ● – Affected male
(4) □ – Unaffected male

133. Choose the **wrong** match

(1)	F. Meischer	–	First identified DNA in nucleus
(2)	Wilkins & Franklin	–	X-ray diffraction data of DNA
(3)	Altman	–	Termed nuclein to DNA
(4)	E. Chargaff	–	Gave ratios between nitrogenous bases A and T and G and C for dsDNA

(1) (1) (2) (2)
(3) (3) (4) (4)

134. Heterochromatin region is
(1) Genetically active
(2) Rich in loosely packed DNA
(3) Genetically inactive
(4) Highly condensed with tightly packed RNA

135. Which of the following is not a symptom shown by individual inflicted with down's syndrome?
(1) Congenital heart disease (2) Mental retardness
(3) Gynaecomastia (4) Furrowed tongue

Zoology

136. The effect of GnRH produced by the hypothalamus directly stimulates the
(1) Synthesis and secretion of gonadotrophins
(2) Secretion of milk in mammary glands
(3) Synthesis of carbohydrates
(4) Fetal ejection reflex

137. In a resting neuron,
(1) Potential difference does not exist across axolemma as ion concentration is same in ICF and ECF
(2) Neural membrane is selectively permeable for all negatively charged proteins
(3) Permeability for potassium is higher than that for sodium ions
(4) Extracellular surface of axolemma is negatively charged

138. Select the **incorrect** option w.r.t. parathyroid gland and its secretion.
(1) PTH is hypercalcemic and can maintain Ca^{2+} homeostasis
(2) Parathyroid hormone is a steroid hormone that gets activated when there is low Ca^{2+} level in blood
(3) PTH may lead to activation of osteoclasts
(4) PTH stimulates reabsorption of Ca^{2+} by the renal tubules

139. Which one is **not** related with action of glucagon?
(1) Gluconeogenesis (2) Glycogenolysis
(3) Deamination (4) Glycogenesis

140. A : Polydipsia and ketonuria are symptoms observed in patients suffering from diabetes insipidus.
R : Fall in ANF is the major cause for occurrence of this disease.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

141. Aqueduct of Sylvius is found between
(1) Paracoel and lateral ventricles
(2) Third and fourth ventricles
(3) Lateral ventricles and third ventricles
(4) Diocoel and lateral ventricles

142. The normal rhythm of sleep wake cycles are maintained by a hormone which is secreted by a gland located
(1) On dorsal side of forebrain
(2) In the neck
(3) In the mediastinum
(4) Above the kidneys

143. Among the following hormones that does not show tropic effect is
(1) FSH, LH (2) TSH, ACTH
(3) Oxytocin, vasopressin (4) LH, TSH

144. Select the option which is **not** a direct function of gonadotropins.
(1) Androgen secretion from testis
(2) Maintenance of corpus luteum
(3) Ovulation
(4) Maintenance of secondary sexual characters

145. Oxytocin, a peptide hormone from hypothalamus, has all the following effects **except**
A. Stimulates vigorous contractions of the uterus at the time of child birth
B. Its hyposecretion causes Addison's disease
C. Stimulates milk ejection from mammary glands
D. Its synthetic form is often given to induce labor
(1) A & C (2) B only
(3) B & D (4) C & D

146. Presence of which of the following is **not** an exclusive feature of a neuron?
(1) Plasma membrane (2) Nissl's granules
(3) Dendrites (4) Axon

147. All of the following hormones influence the glucose metabolism, **except**
(1) Epinephrine (2) Cortisol
(3) Insulin (4) Aldosterone

148. Hormone that influences metabolism, pigmentation, the menstrual cycle as well as our defense capability, is
(1) MSH (2) Melanin
(3) Melatonin (4) FSH

149. Select the **mismatch**.
(1) ANF – Maintains blood volume
(2) Parathormone – Hypercalcemic hormone
(3) Thyrocalcitonin – Hypocalcemic hormone
(4) Progesterone – Follicular growth

150. Hormones are ___A___ chemicals which act as ___B___ messengers and are produced in trace amounts.
Choose the **correct** option for A and B.

	A	B
(1)	Nutrient	Intercellular
(2)	Non-nutrient	Intracellular
(3)	Non-nutrient	Intercellular
(4)	Inorganic	Intracellular

(1) (1) (2) (2)
(3) (3) (4) (4)

151. Cortisol can perform all the following functions **except**
(1) Lipolysis and proteolysis
(2) Gluconeogenesis
(3) Immunosuppressive hormone and stimulates erythropoiesis
(4) Inflammatory response

152. Secondary messenger is not required by hormone
(1) PTH (2) TCT
(3) Estrogen (4) FSH

153. Hypersecretion of GH can cause all of the following, **except**
(1) Acromegaly (2) Dwarfism
(3) Hyperglycemia (4) Gigantism

154. In a hyperpolarised neuron, the potential difference across axolemma can be
(1) +30 mV (2) –55 mV
(3) –90 mV (4) –70 mV

155. Read the following statements and choose the **incorrect** one.
(1) Vasopressin is produced by hypothalamus and stored in pituitary
(2) Deficiency of growth hormone can result in proportional dwarfism unlike acromicria
(3) Light promotes melatonin formation
(4) Adrenal gland is ectomesodermal in origin

156. Hypothalamus has centres for all of the following, **except**
(1) Hunger (2) Thirst
(3) Gastric secretions (4) Thermoregulation

157. Hormone responsible for formation of RBC in human body is produced by
(1) Heart (2) Liver
(3) Bone marrow (4) Kidney

158. Glucagon and insulin hormone can be distinguished based on
(1) Location of receptors *i.e.* intracellular or extracellular
(2) Their source gland
(3) Hepatocytes as target cells
(4) Their role in the mechanism of maintaining glucose homeostasis

159. Which of the following is **not** the action of sympathetic nervous system?
(1) Dilates pupil (2) Increases heart rate
(3) Increases salivation (4) Relaxes urinary bladder

160. The hypothalamic regulatory factors are carried into _____ through hypophyseal portal vein.
Select the **correct** option to fill in the blank.
(1) Pineal gland (2) Adenohypophysis
(3) Neurohypophysis (4) Pars nervosa

161. Cerebral aqueduct is a canal which connects the cavity of diencephalon to cavity of medulla oblongata and passes through
(1) Cerebrum (2) Midbrain
(3) Cerebellum (4) Hypothalamus

162. Following are the statements about polarised state as well as process of impulse conduction through a nerve fibre.
a. Neurons are non-excitable cells because their membranes are in a polarised state.
b. Polarised state of a nerve fibre is due to the difference in permeability of Na^+ and K^+.
c. The electrical potential difference across the resting plasma membrane is called the action potential.
d. Resting potential of axolemma is restored at the site of excitation by rapid efflux of K^+.
Choose the **correct** answer from the options given below.
(1) a and b are correct (2) b and c are correct
(3) c and d are correct (4) b and d are correct

163. A patient is suffering from tumor of zona glomerulosa of adrenal cortex that leads to the abnormal level of secretion of aldosterone. Select the **true** feature regarding urinary composition of the patient.
(1) High Na^+ and low K^+ concentration
(2) Low Na^+ and high K^+ concentration
(3) High Na^+ and high K^+ concentration
(4) Low Na^+ and low K^+ concentration

164. Choose the **incorrect** match.
(1) Hypothalamus – Centre of neurohormones
(2) Calcitonin – Deficiency causes tetany
(3) Thyroid hormone – Hypersecretion causes Graves' disease
(4) Epinephrine – Derivative of tyrosine

165. Which of the following endocrine glands stores and releases neurogenic hormones?
(1) Thyroid gland (2) Neurohypophysis
(3) Adenohypophysis (4) Parathyroid glands

166. Which of the following is **not** a function of LH?
(1) Induces ovulation of Graafian follicle
(2) Maintains corpus luteum
(3) Induces Leydig cells to secrete androgens
(4) Acts on cells of Sertoli for spermatogenesis

167. Select the **incorrect** statement about atrial natriuretic factor (ANF).
(1) It is a peptide hormone acting as vasodilator.
(2) It is synthesized from atrial wall of heart.
(3) It increases blood pressure to increase the GFR.
(4) It works antagonistically to renin-angiotensin-aldosterone system.

168. All of the following hormones are derived from amino acid tyrosine **except**
(1) Thyroxine
(2) Epinephrine
(3) Melatonin
(4) Norepinephrine

169. Corpus callosum, a tract of nerve fibres connects
(1) Two cerebellar hemispheres
(2) Two cerebral hemispheres
(3) Two superior colliculi
(4) Two inferior colliculi

170. Secretion of β-cells of pancreas does **not** promote
(1) Synthesis of fats from glucose in adipose tissue.
(2) Uptake of glucose by liver cells
(3) Metabolic breakdown of fat
(4) Utilisation of glucose by cells

171. The disease that is **not** caused by hypothyroidism is
(1) Exopthalmic goitre
(2) Cretinism
(3) Simple goitre
(4) Myxedema

172. Which of the following is not a function of hypothalamus?
(1) It has thermoregulating centre
(2) It has hunger and thirst centre
(3) It secretes neurohormones
(4) It controls respiration, cardiovascular reflexes and gastric secretions

173. Select the **incorrect** statement about electrical synapses.
(1) They are rare in our system
(2) Impulse transmission is faster as compared to a chemical synapse
(3) Pre and post-synaptic membranes possess a wide gap between them
(4) Neurotransmitters are absent at these synapses

174. Select a pair of hormones having synergistic effects
(1) TCT and PTH w.r.t blood calcium levels
(2) Glucagon and Insulin w.r.t blood glucose levels
(3) Epinephrine and norepinephrine w.r.t lipolysis
(4) ANF and ADH w.r.t blood pressure

175. The $Na^+ - K^+$ pumps transport how many Na^+ and K^+ ions per ATP spent for maintaining the polarised state when the neuron is **not** conducting any impulse?
(1) 2 Na^+ and 3 K^+
(2) Na^+ and 3 K^+
(3) 3 Na^+ and 2 K^+
(4) 2 Na^+ and K^+

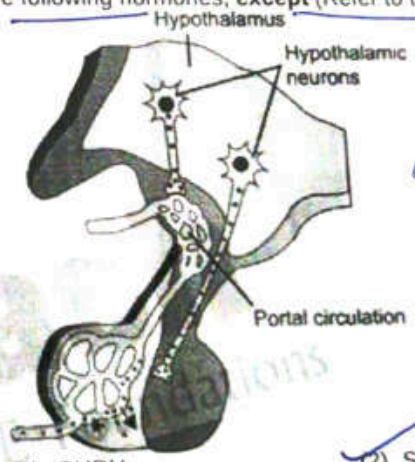
176. Nissl's granules are present in
(1) Cyton of human bipolar neuron
(2) Axon of multipolar neuron
(3) Schwann cells of myelinated neuron
(4) Synaptic knob of pseudounipolar neuron

177. Corpora quadrigemina is a part of
(1) Forebrain
(2) Hindbrain
(3) Midbrain
(4) Cerebrum

178. Which hormone acts on exocrine part of pancreas and stimulates secretion of water and bicarbonate ions?
(1) Gastrin
(2) Secretin
(3) CCK
(4) GIP

179. **Assertion (A):** The cerebral cortex is referred to as grey matter due to its greyish appearance.
Reason (R): The neuronal cell bodies are concentrated in the cerebral cortex.
In the light of above statements, select the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

180. If we draw out some blood from the hypothalamo hypophyseal portal vein, it will be found to contain all of the following hormones, **except** (Refer to the figure)



(1) GHRH
(2) Somatostatin
(3) PRH
(4) Somatotropin